

4. Collaboration Diagram.

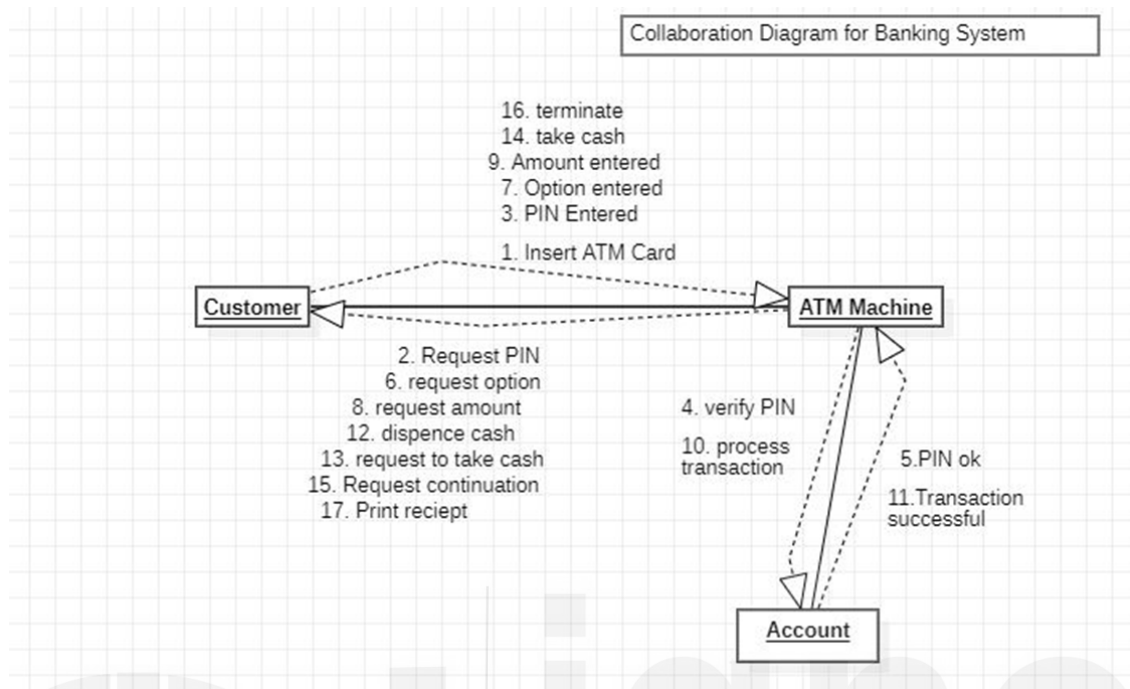


Figure 1.13: Collaboration Diagram for Banking System

5. Deployment Diagram.

Deployment Diagram For ATM System

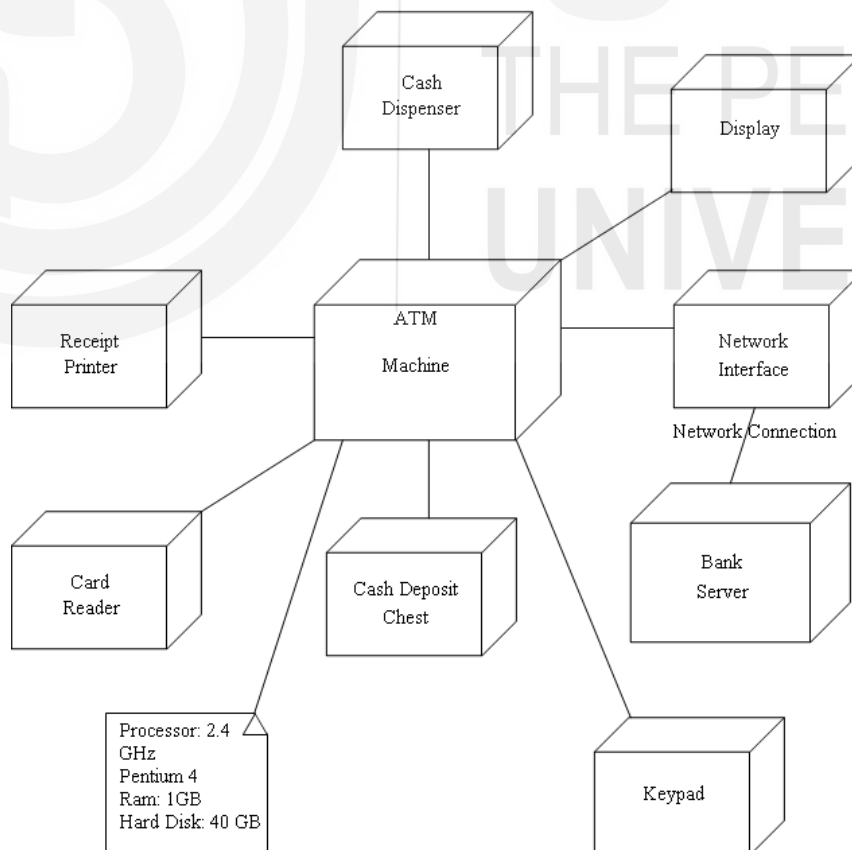


Figure 1.14: Deployment Diagram

1.5 SESSION WISE LIST OF LAB EXERCISE

This section contains a list of exercises that you should perform during your lab sessions. You need to prepare a draft solution prior to start of the session and implement it during the designated lab session. **While solving/answering each of the following exercises, you should write problems description in about 300-500 words and also list the basic assumptions related to the problem. This will help you to limit your solution within the specified scope and will also help the evaluator in evaluating your answers/solutions.**

Session-1

1. Draw Class Diagram for Online Shopping System (First, list the assumptions of the shopping system and then identify classes, and then draw the diagram).
2. Draw Class Diagram for Online Examination System.
3. Draw Object Diagram for Online Banking System.

Session-2

4. Draw Use Case Diagram for Online Shopping System.
5. Draw Use Case Diagram for Library Management System.

Session-3

6. Draw Sequence Diagram for Online Shopping System.
7. Draw Sequence Diagram for Online Examination System.
8. Draw a Sequence diagram for Employee Management System.

Session-4

9. Draw Collaboration Diagram for Student Registration Process in Masters Program of a University.
10. Draw Collaboration Diagram for Online Banking System.
11. Draw a Collaboration diagram for Employee Management System.

Session-5

12. Activity Diagram Online Banking System.
13. Draw Activity Diagram for Online Examination System.

Session-6

14. State Chart Diagram for Online Examination System.
15. State Chart Diagram for Online Fund Transfer through Netbanking Banking System.

Session-7

16. Draw Component Diagram for Online Examination System.
17. Draw Component Diagram for Order Processing Application.
18. Diagram Deployment Diagram for Online Student Admission System.

Session-8

19. Implement the following Class Diagram in C++/Java.

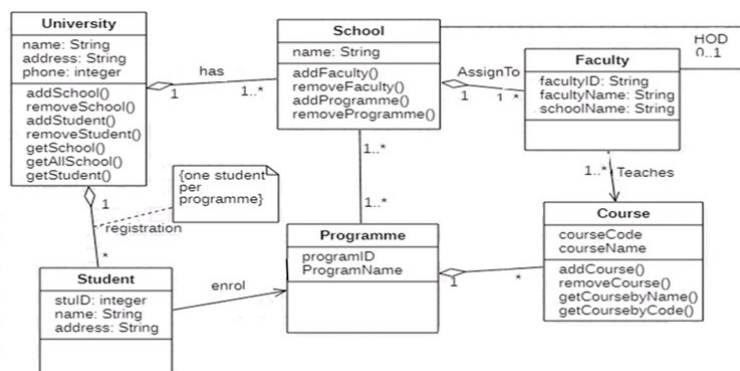


Figure 1.15: Class Diagram for Student Registration

20. Implement the Class Diagram of figure 1.18, in C++ or Java.

Session-9

21. Implement the following Associations using C++/Java.

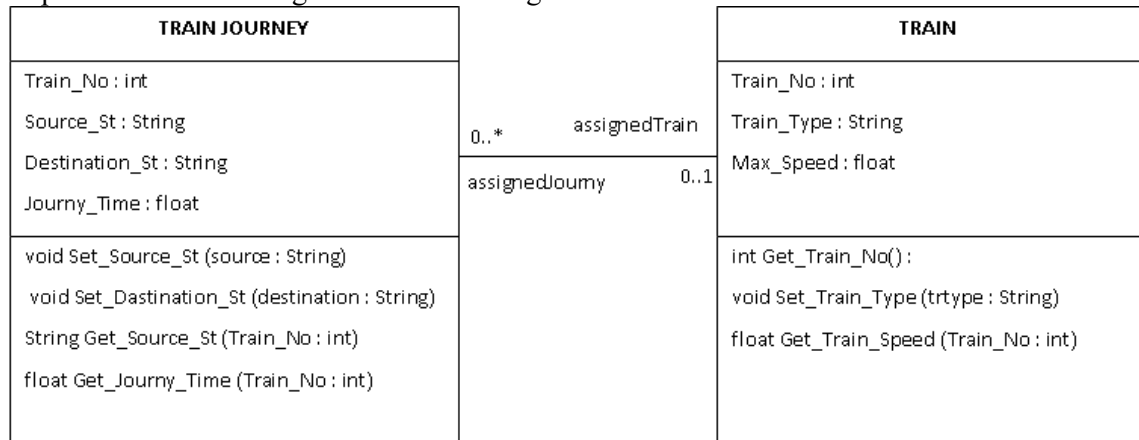


Figure1.16: Train-Journey Association

22. Implement the following Associations using C++/Java.

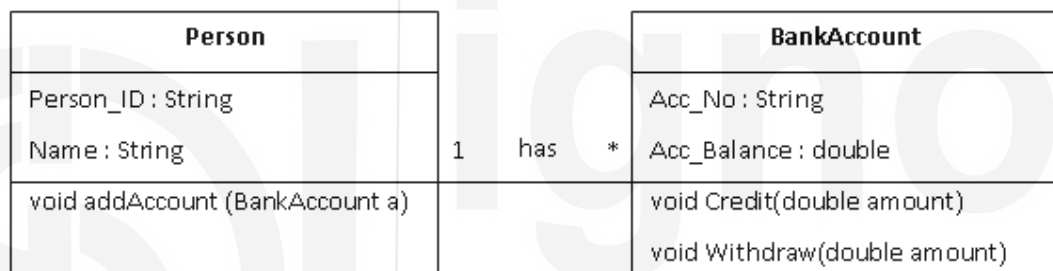


Figure 1.17: Person and Bank Account

Session-10

23. Do mapping of the following Classes into database tables

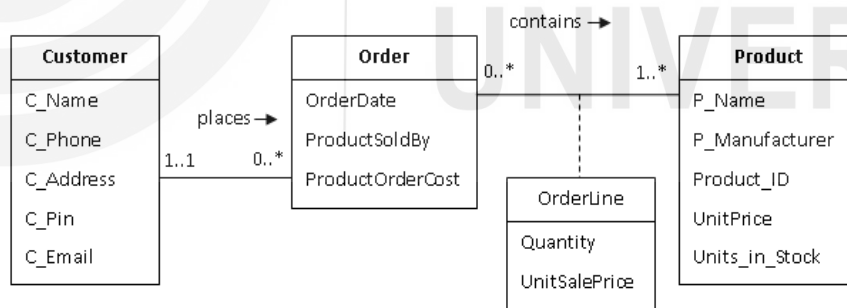


Figure1.18: Customer Order Association Class

1.6 REFERENCES/FURTHER READINGS

- Grady Booch, James Rumbaugh and Ivar Jacobson, “The Unified Modeling Language User Guide”, Pearson Education, 2004.
- <https://argouml.en.softonic.com/>
- <https://argouml-tigris-org.github.io/tigris/argouml/>
- <https://staruml.io/>
- <https://docs.staruml.io/>
- <https://docs.staruml.io/user-guide/editing-elements>